

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 2.4: Data Types and Data Problems

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Data, Data, Data

- In all the examples discussed, we dealt with data as input for machine learning algorithms: numbers, images, ...
- Many of the pitfalls in using AI have to do with the data. "Garbage in, garbage out".

In the following, we will focus intensively on data. What kind of data is there? How can we describe it? And how do we need to handle it to avoid errors?

Numbers

Different Types of Numbers

- Numbers come in two types:
 - **Discrete:** Integers. Examples: number of people in a room, days until Christmas, calls to the phone hotline.

*Example:
Information about
different wines.*

AGE	RATING	SUGAR
12	1	1.3
13	1	1.7
1	2	1.5
8	3	3.3
3	4	3.0
2	3	2.1

Different Types of Numbers

- Numbers come in two types:
 - **Discrete:** Integers. Examples: number of people in a room, days until Christmas, calls to the phone hotline.
 - **Continuous:** Decimal numbers (floats). Examples: weight of a person, distance to Paris, percentage of women in a lecture, temperature.

*Example:
Information about
different wines.*

AGE	RATING	SUGAR
12	1	1.3
13	1	1.7
1	2	1.5
8	3	3.3
3	4	3.0
2	3	2.1

Tables

Often also called "data frame", "data table", or "matrix".

column

all values of a given attribute

AGE	RATING	SUGAR
12	1	1.3
13	1	1.7
1	2	1.5
8	3	3.3
3	4	3.0
2	3	2.1

row

all values

of an

observation

Problems with Numbers

- Missing values can be blanks in the data, but also marked with placeholders like "-1" or "NA".

*Example:
Information about
different wines.*

AGE	RATING	SUGAR
-1	1	1.3
13	1	NA
1		1.5
8		3.3
3	4	3.0
-1	3	NA

Problems with Numbers

- **Missing values** can be blanks in the data, but also marked with **placeholders** like "-1" or "NA".
- **Outliers**: values that are far outside the expected range.

*Example:
Information about
different wines.*

AGE	RATING	SUGAR
12	1	0.001
13	1	1.7
1	2	1.5
8	3	3.3
1998	4	3.0
2	3	2.1

Problems with Numbers

- **Missing values** can be blanks in the data, but also marked with **placeholders** like "-1" or "NA".
- **Outliers**: values that are far outside the expected range.
- **Forbidden values**: values that are outside the **allowed** range.

*Example:
Information about
different wines.*

AGE	RATING	SUGAR
12	1	1.3
13	1	1.7
1	2	1.5
-10	3	3.3
3	4	3
2	3.3	2.1

Finding Problems with Numbers

- **Check assumptions about numbers**
 - Which values are allowed?
 - Which value ranges are expected?
 - How are missing values encoded?
- **Descriptive Statistics**
 - **Minimum:** the smallest number
 - **Maximum:** the largest number
 - **Median:** which number is exactly in the middle if we order all values from smallest to largest? Example: [1, 1, 3, 5, 6, 6, 13]
 - **Mean:** sum all values and divide them by the count of values
 - **Mode:** the most frequent number

Fixing Problems with Numbers

- **Missing Values**

- Remove columns or rows with missing values.
- Imputation: replace missing values with mean, median, or mode.

- **Outliers**

- Find out where the values come from.
- Replace with minimum / maximum or missing value.

- **Forbidden Values**

- Find out where the values come from.
- Replace with missing values or impute.

Classes / Categories

Different Types of Classes

- **Category:** Can have more than two values. Examples: color, gender, place of residence, ethnicity

*Example:
Information about
different wines.*

COLOR	AVAILABLE	RATING
red	yes	1
white	yes	1
white	no	2
red	no	3
white	yes	4
red	no	3

Different Types of Classes

- **Category:** Can have more than two values. Examples: color, gender, place of residence, ethnicity, quality.

*Example:
Information about
different wines.*

COLOR	AVAILABLE	RATING
red	yes	1
white	yes	1
white	no	2
red	no	3
white	yes	4
red	no	3

Different Types of Classes

- **Category:** Can have more than two values. Examples: color, gender, place of residence, ethnicity, quality.
- **Boolean:** "yes" or "no", often "1" or "0".

*Example:
Information about
different wines.*

COLOR	AVAILABLE	RATING
red	yes	1
white	yes	1
white	no	2
red	no	3
white	yes	4
red	no	3

How do we "Compute" with Classes?

- **Boolean values** (binary attributes): translate into "0" and "1".

*Example:
Information about
different wines.*

AVAILABLE	COLOR
1	red
1	white
0	white
0	red
1	white
0	red

How do we "Compute" with Classes?

- **Boolean values** (binary attributes): translate into "0" and "1".
- **Categories with two possible values**: translate into boolean values.

*Example:
Information about
different wines.*

AVAILABLE	RED
1	1
1	0
0	0
0	1
1	0
0	1

How do we "Compute" with Classes?

- **Boolean values** (binary attributes): translate into "0" and "1".
- **Categories with two possible values**: translate into boolean values.
- **Categories with more than two possible values**

*Example:
Information about
different wines.*

AVAILABLE	COLOR
1	red
1	white
0	white
0	red
1	white
0	rosé

How do we "Compute" with Classes?

- **Boolean values** (binary attributes): translate into "0" and "1".
- **Categories with two possible values**: translate into boolean values.
- **Categories with more than two possible values**: split into multiple columns ("one-hot encoding").

*Example:
Information about
different wines.*

AVAILABLE	RED	WHITE	ROSÈ
1	1	0	0
1	0	1	0
0	0	1	0
0	1	0	0
1	0	1	0
0	0	0	1

Images

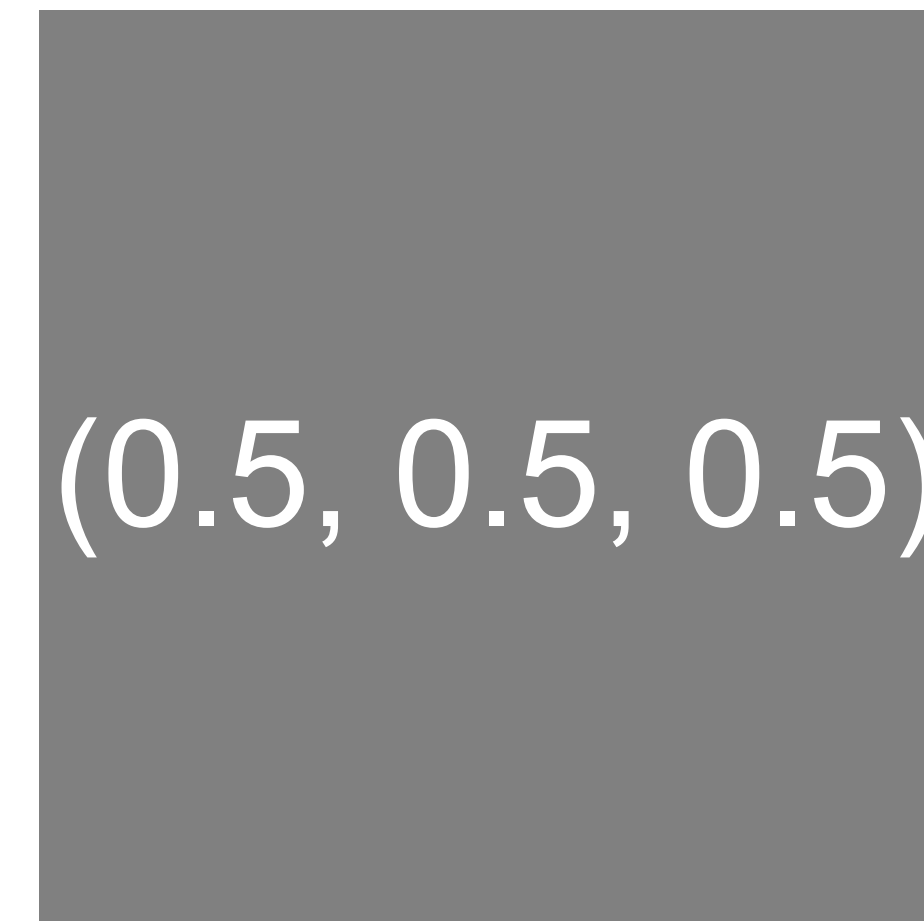
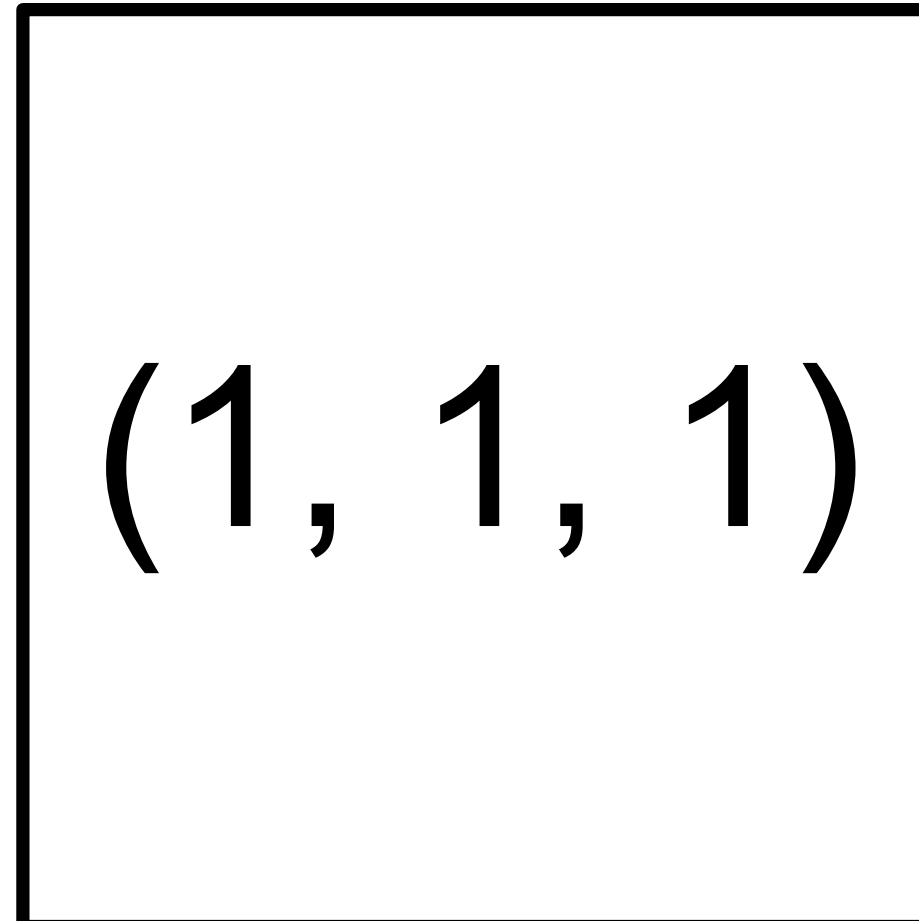
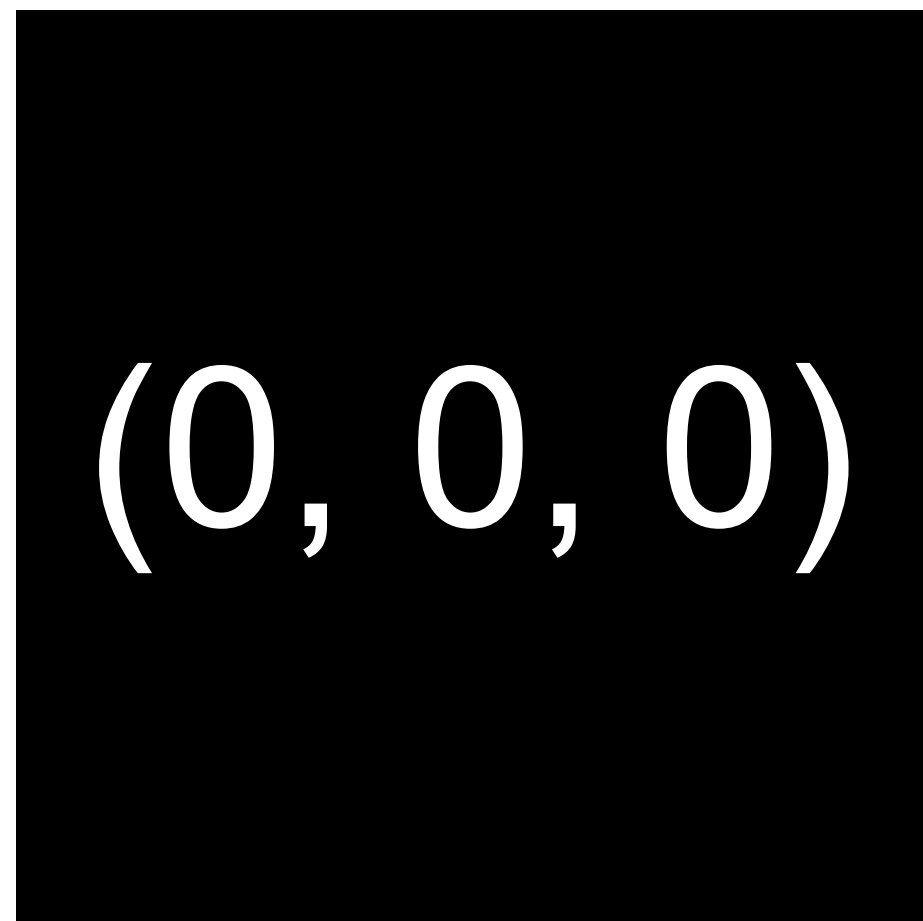
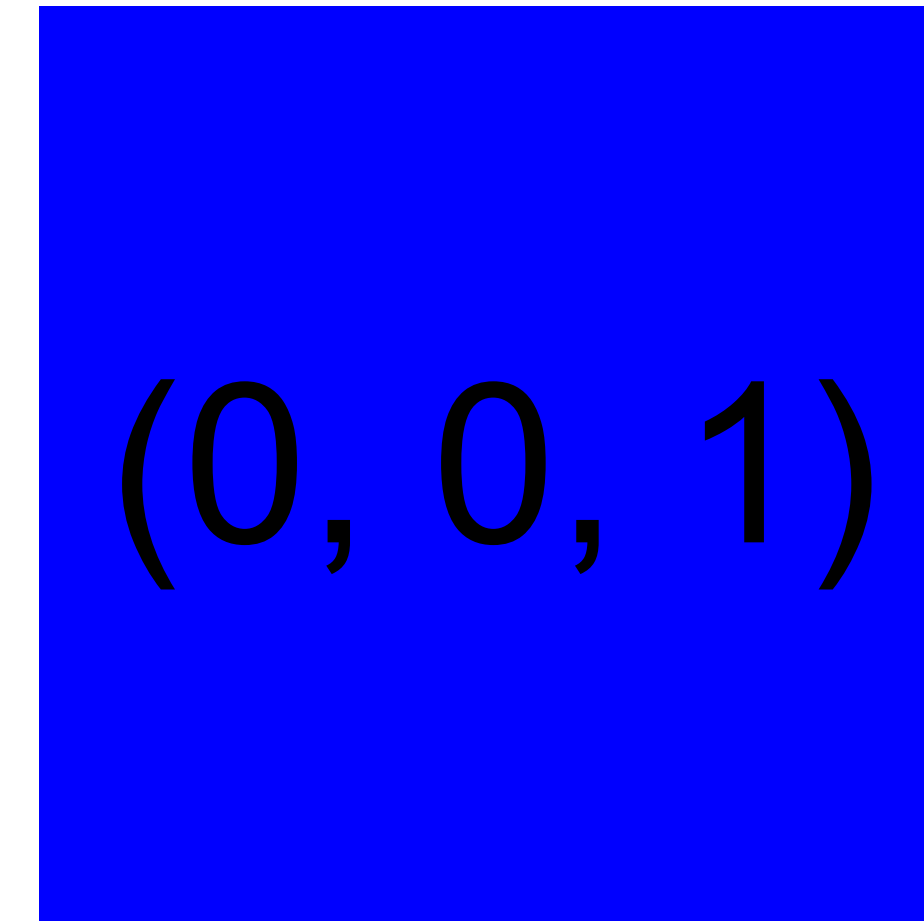
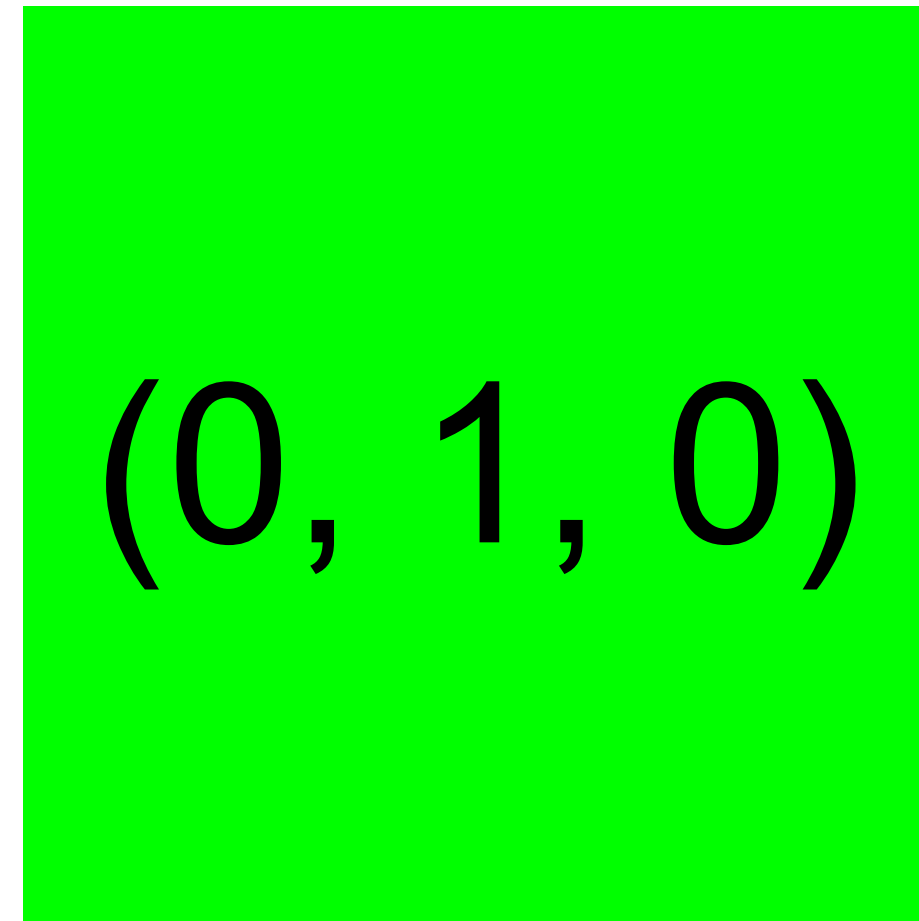
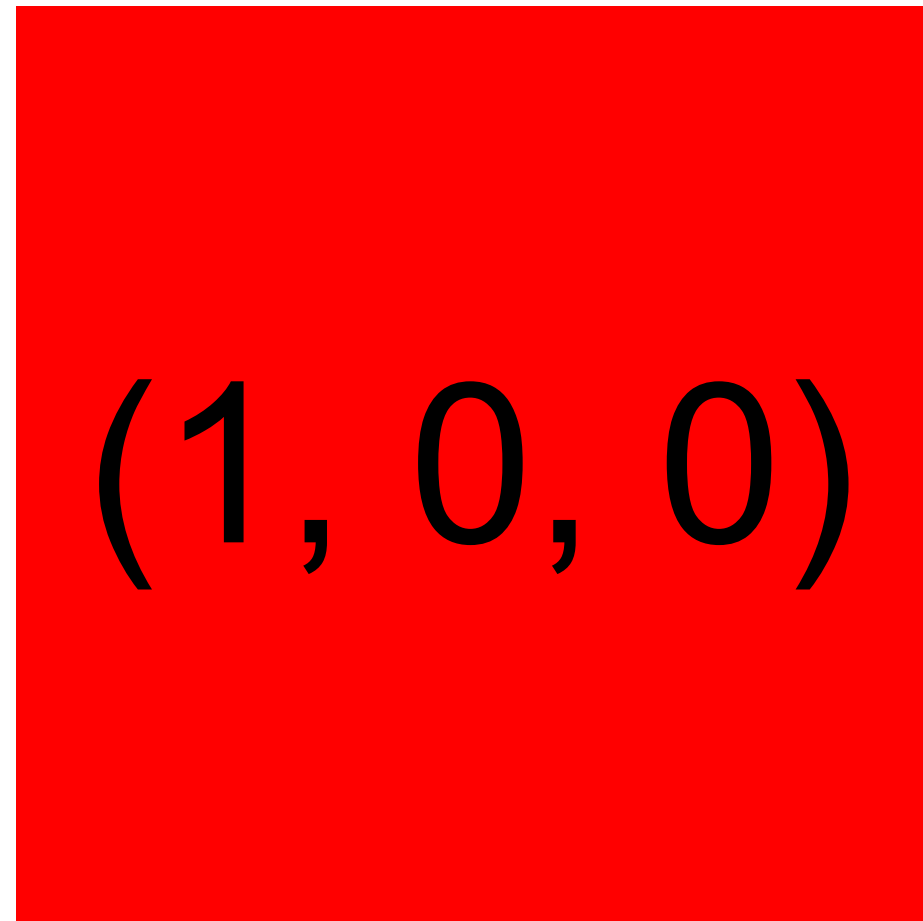
Images consist of Pixels

Pixels (picture elements) already contain numbers – the color values of the pixel. Images already have rows and columns. Actually, images are already tables.

	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)	(0,6)	(0,7)	(0,8)
(0,0)	1	1	1	1	1	1	1	1
(1,0)	1	1	1	1	0	0	1	1
(2,0)	1	1	1	0	0	0	1	1
(3,0)	1	1	0	0	0	0	1	1
(4,0)	1	0	0	1	0	0	1	1
(5,0)	1	0	1	1	0	0	1	1
(6,0)	1	1	1	1	0	0	1	1
(7,0)	1	1	1	1	0	0	1	1
(8,0)	1	1	1	1	0	0	1	1
(9,0)	1	1	1	1	1	1	1	1

(0,1)	(0,2)	(0,3)	(0,4)	(0,5)	(0,6)	(0,7)	(0,8)
1	1	1	1	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	0	0	0	1	1
1	1	0	0	0	0	1	1
1	0	0	1	0	0	1	1
1	0	1	1	0	0	1	1
1	1	1	1	0	0	1	1
1	1	1	1	0	0	1	1
1	1	1	1	0	0	1	1
1	1	1	1	1	1	1	1

Images with Colors



Text

Representing Text as Numbers

I take the
dog
outside

	I
I	1
take	0
the	0
dog	0
outside	0

Representing Text as Numbers

I take the
dog
outside

	I	take	the	dog	outside
I	1	0	0	0	0
take	0	1	0	0	0
the	0	0	1	0	0
dog	0	0	0	1	0
outside	0	0	0	0	1

Representing Text as Numbers

I take the
dog
outside

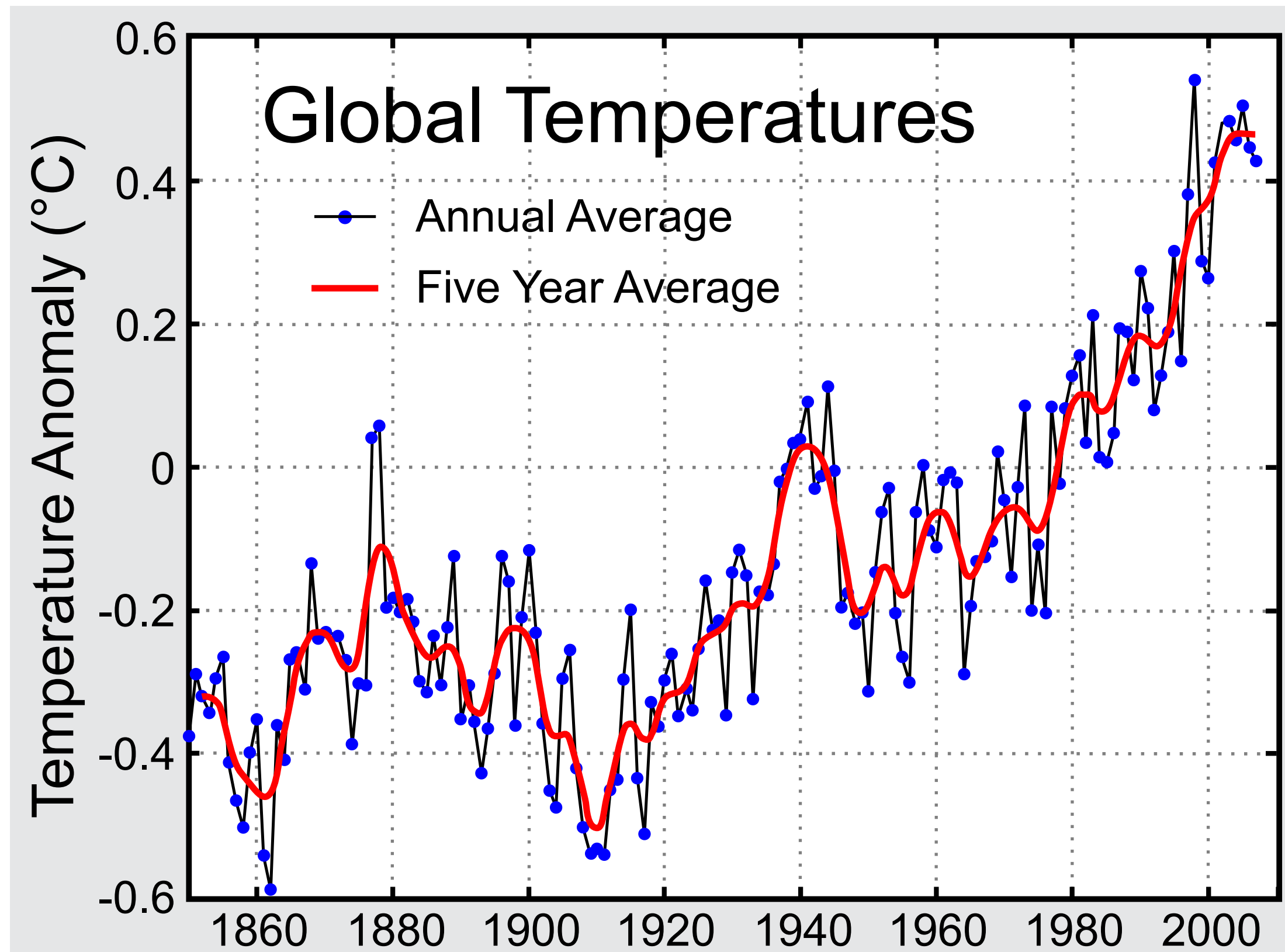
	I	take	the	dog	outside
I	1	0	0	0	0
take	0	1	0	0	0
the	0	0	1	0	0
dog	0	0	0	1	0
outside	0	0	0	0	1
mouse	0	0	0	0	0
cat	0	0	0	0	0
house	0	0	0	0	0
...	...	...	...	...	...
you	0	0	0	0	0

- There are around 600,000 words in the Oxford English Dictionary.
- This idea quickly leads to huge tables.
- More on this in Lecture 4!

Further Data Types

Time Series

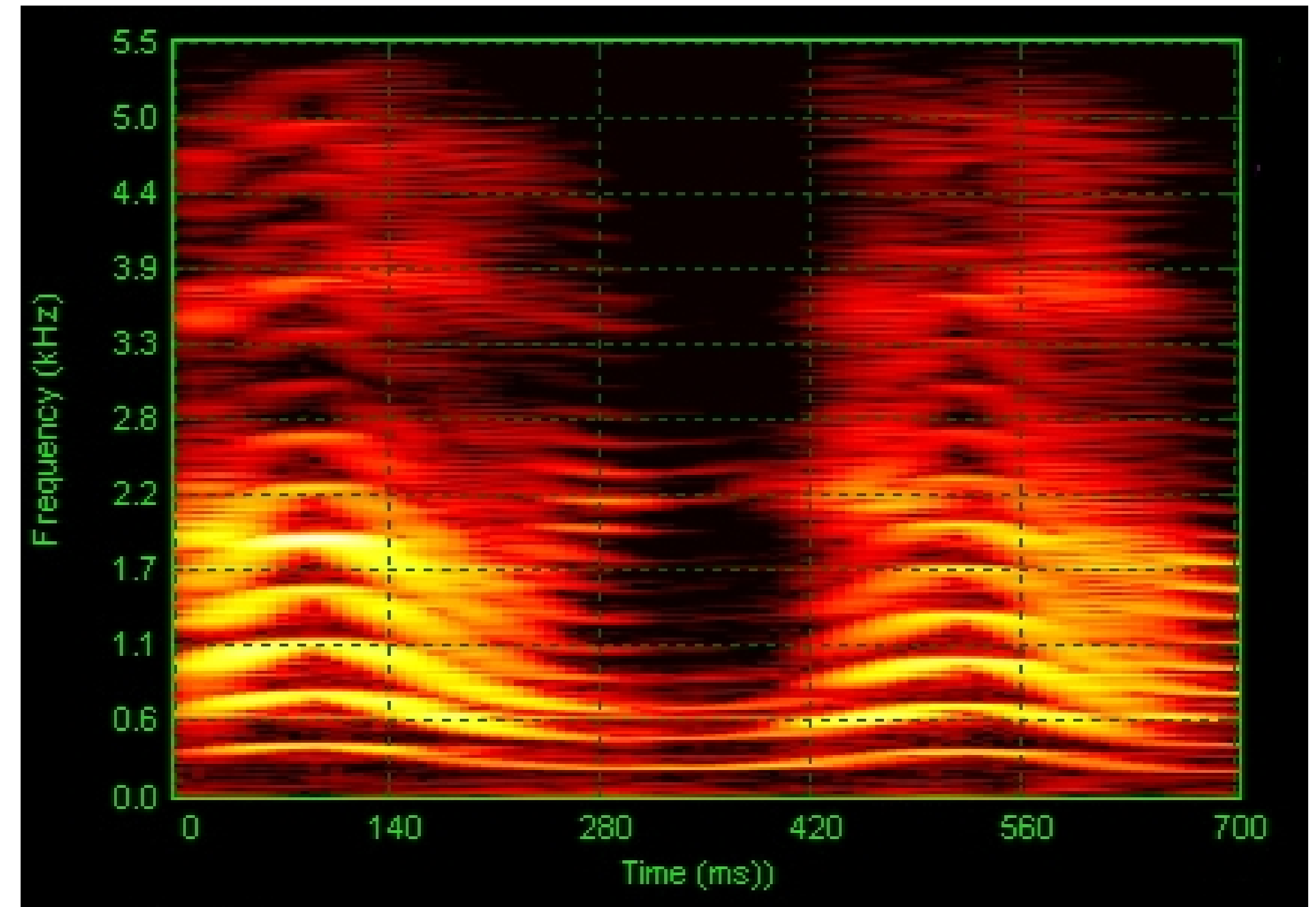
Individual numbers recorded at regular time intervals.



Temperature curve since 1850. Source: [Wikipedia](#), CC BY-SA 3.0

Sound

Many numbers (frequencies) recorded at regular time intervals.



Spectrogram of the song "Oh No Girl". Source: [Wikipedia](#), CC BY-SA 3.0

Summary

- To make data usable for machine learning, we need to convert it into **numbers in tables**.
- There are different **data types**, all of which can be represented in tabular form in one way or another.
 - Numbers: discrete and continuous
 - Classes: boolean (binary attributes) and categories
 - Images: black and white, and color
 - Text
 - Time series
- To discover and fix **data problems**, it's helpful to look at **descriptive** statistics (minimum, maximum, mode, median, mean).

Exercise (5 min)

Think back to the supervised learning application from your daily life that you considered in the last lecture.

- What data could you collect to train your algorithm?
- What problems could arise with this data?
- How could you fix these problems?